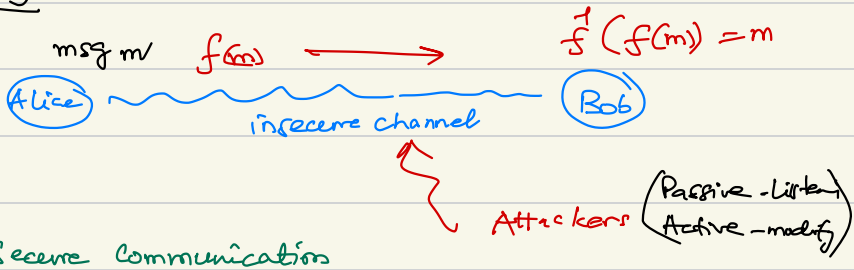
PQC

IIIT Pune

Cryptography :



Goal: Secure communication

— Confidentiality of the message

— Integrity

Basic requirement: f must be invertible

Alice knows f
Bob knows f^{-1}
} Attacker does not know f^{-1}.

Kerckhoff's Principle :

"La Cryphographie Militaire"

"System" is known to the attacker.

Symmetric Key Cryptography :

there must be a secret shared between A & B

— Which is not known to the attacker

Secret

(i) Algorithm — hiding f^{-1} from the attacker (Steganography)

(ii) Data —

Kerckhoff's Principle :

All algorithms used by A & B are known to the attacker, only secret data is not known

Secret key – the secret data shared between A & B.

Secret Communication:

$\{0,1\}^n$ = n-bit strings

(i) Key Generation Algorithm (1^n) → key as output

n = security parameter

Should run in time polynomial in n.

Efficient algorithm:

Algorithm should run in time $O(n^k)$

where n = length of the input & k = some constant

(ii) **Encryption.**

$Enc(k, msg) = ciphertext = c$

(iii) **Decryption**

$Dec(k, c) = m$

For correctness: we need:

$\forall$ key k, $\forall$ msg m, $\forall$ ciphertext c

$$Dec(k_2, Enc(k_1, m)) = m$$

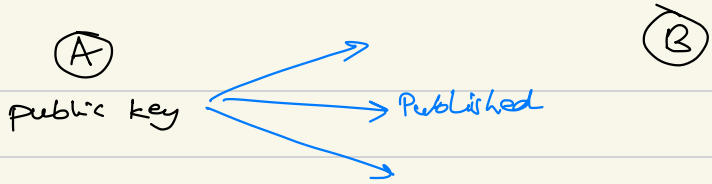
Symmetric key Crypto.

Public key Cryptography

$$Dec(k_2, Enc(k_1, m)) = m$$

$k_1 \neq k_2$

k_2: Decryption key; k_1: Encryption key



Defining Security:

Attempt 1: attacker doesn't learn m when given $Enc(pk, m) = c$

Consider an attacker:

Pick a random msg m

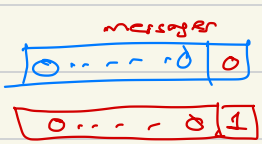
$c = Enc(pk, m)$

Claim: that you know m for c

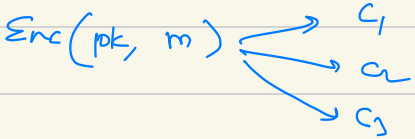
Attempt 2: should include randomization in the message generation

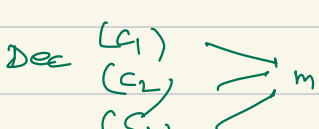
Suppose input is only 1 bit → 0 / 1

$c = Enc(pk, m)$

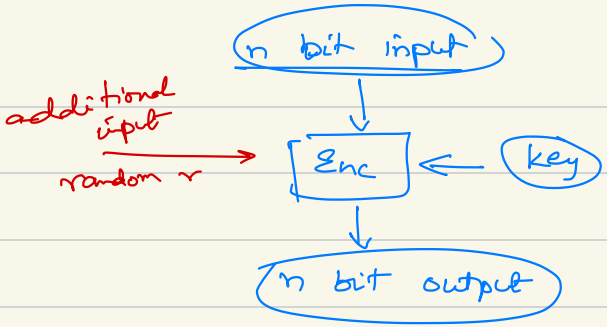


Encryption algo. must be randomized





Deterministic



$Enc(m_1) \rightarrow c$
$Enc(m_2) \rightarrow c$
X

$Dec(k, r, c)$
↓
$Dec(k, c)$

c = randomized
= (random part, ...)

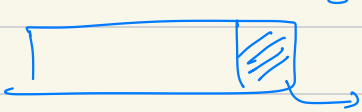
There are methods, where you can eliminate "random part" from the ciphertext

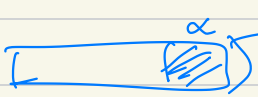
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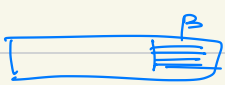
$Enc(key, \underset{letter}{appointment}) = ciphertext$

Attacker has (m_1, c_1)
& $c_2, c_3, c_4, \ldots$

Goal: to get some information from c_2



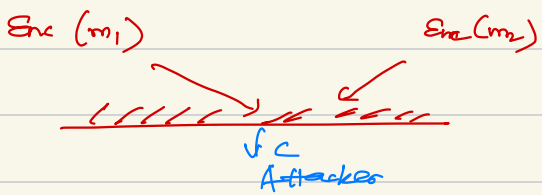
$c_2 = Enc($  $)$

$c_1 = Enc($  $)$

Goal: given c_1, c_2, find if $(\alpha > \beta)$ or not

Idea: Given a target ciphertext c^* attacker should not be able to reveal any "non-trivial" information about m.

Indistinguishability



Power of the attacker

① Only ciphertexts are known: $c_1, c_2, c_3 \ldots$
- gets a target c^* to decrypt
- or distinguish between two messages given a target ciphertext

② Known plaintext attack

$(m_1, c_1), (m_2, c_2), \ldots (m_t, c_t)$

target c^* or distinguish between Enc of m_1 & m_2

③ Chosen Plaintext Attack

attacker chooses $m_1, m_2, \ldots$

$\downarrow c_1 \quad \downarrow c_2 \quad \ldots$

④ Chosen Cipher Attack (CCA)

- attacker can ask for Enc (m) for any m
- " " " Dec (c) for any c

- finally attacker chooses m_0, m_1

$b \leftarrow \{0,1\}$ then the encryptor produces a ciphertext c for message (m_b)

attacker guesses which one was used to produce c^*.

$$A(1^n, c^*) \rightarrow b'$$

$$Pr[A \text{ produces } b' \text{ matching with } b] = \frac{1}{2} + \epsilon$$

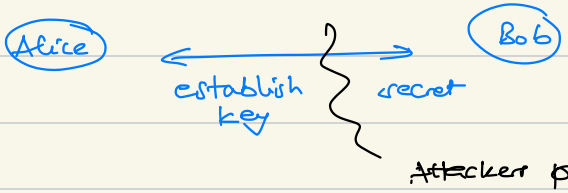
$$\epsilon = \frac{1}{2^n} \quad \text{or sometimes similar}$$

$$= negl(n)$$

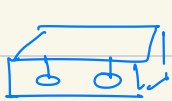
Notion: IND – CCA

Indistinguishability under Chosen Ciphertext Attack

Designing Enc. schemes of this type:

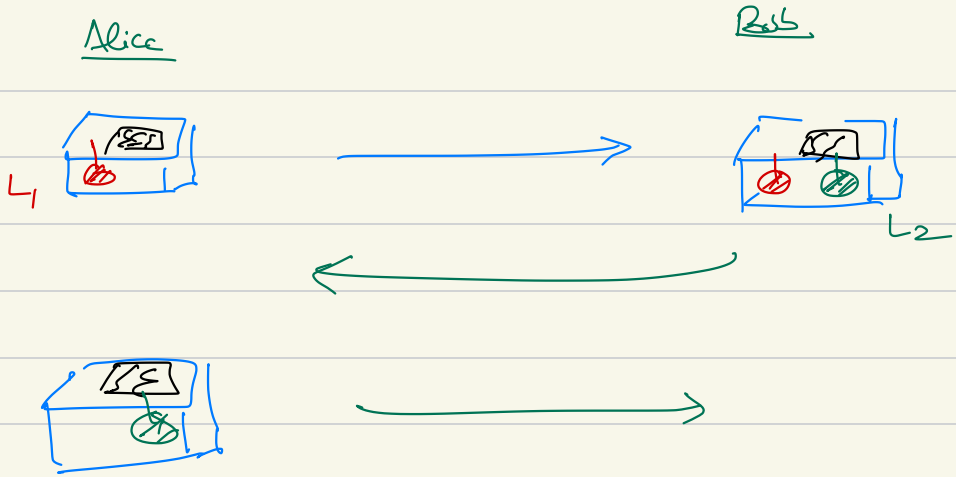


Key Exchange:

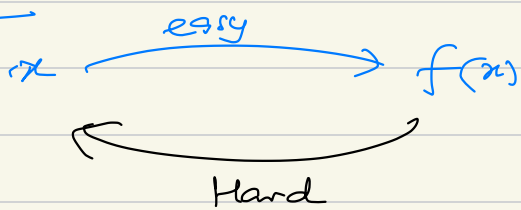


Assumption: trunk can't be broken

we have locks which can't be broken



One-Way function



Group: Set, + operation

$(G, *)$ → binary operation

① **Closure:** $\forall\ g_1, g_2 \in G$

$g_1 * g_2 \in G$

② **Associativity** $\forall\ g_1, g_2, g_3 \in G$

$g_1 * (g_2 * g_3) = (g_1 * g_2) * g_3$

(Notice that $g_1 * g_2$ may or may not be = to $g_2 * g_1$)

$\Rightarrow$ Suppose $(g * g) * g = g * (g * g)$

$\Rightarrow$ $g * g * g = g^3$

③ **Identity** : $\exists$ a special element called e

s.t. $\forall \; g \in G$

$g * e = e * g = g$

④ **Inverse** : $\forall \; g \in G$

there exists a g' s.t

$g' * g = e = g * g'$

Additional : if the group has additional

$\forall \; g_1, g_2 \in G$

$g_1 * g_2 = g_2 * g_1$

$\Rightarrow$ Abelian Group

$Z_5 = \{0, 1, 2, 3, 4\}$ set of integers mod 5

op. = $+$ mod 5

$\Rightarrow$ It is a group

$Z_7 = \{0, 1, 2, \ldots 6\}$

op $=$ $\times$ mod 7

Not a group

$Z_7^* = \{1, 2, 3 \ldots 6\}$

op $=$ $\times$ mod 7

We want $a \cdot b \bmod 7 \neq 0 \bmod 7$

Inverse:

$\bar{1}^{-1} = 1$

$\bar{2}^{-1} = 4$

$\bar{3}^{-1} = 5$

$\bar{4}^{-1} = 2$

$\bar{5}^{-1} = 3$

$\bar{6}^{-1} = 6$

Consider $G = \{1, 2, \ldots 9\}$

op: $\times$ mod 10

$\longrightarrow$

$Z_{10}^* = \{1, 3, 7, 9\}$, op = $\times$ mod 10

$$\mathbb{Z}_n^* = \{ x : \quad 1 \le x < n ; \ \gcd(x,n) = 1 \}$$

this is a Group: Multiplicative Group mod n

✓ Closure: if $\gcd(a,n) = 1$, $\gcd(b,n) = 1$ $\Rightarrow \gcd(ab,n) = 1$

$(ab) \bmod n \longrightarrow$

✓ Associativity

✓ Identity $\quad 1 \in G$

Inverse:

$$\mathbb{Z}_n^* = \{ 1, g_1, g_2, \ldots g_t \}$$

Pick any element g from here

Consider $g \cdot \mathbb{Z}_n^* = \{ g, g*g_1, g*g_2, \ldots \}$ ($g*g_j = 1$, j^{th} element $= 1$)

$= \mathbb{Z}_n^*$

How many elements in $g \cdot \mathbb{Z}_n^*$?

Suppose there was some "collision" in this.

$$g * g_i = g * g_j$$

$$\Rightarrow \quad \bar{g} * (g * g_i) = \bar{g} * (g * g_j)$$

$$\Rightarrow \quad (\bar{g} * g) * g_i = (\bar{g} * g) * g_j$$

$$\Rightarrow \quad g_i = g_j$$

Multiplicative group mod n

$$\mathbb{Z}_n^* = \{ x : (1 \le x < n) \ \& \ \gcd(x, n) = 1 \}$$

$$|\mathbb{Z}_n^*| = ? = \phi(n)$$

Euler's totient function

Let p be a prime no.

$$|\mathbb{Z}_p^*| = (p-1)$$

$$|\mathbb{Z}_{p \cdot q}^*| = (p-1)(q-1)$$

↓ RSA

For $n = pq$

$\phi(n) \iff (p, q)$ = factors of n

Why?

Knowing factors → p, q ✓

$(p-1), (q-1)$ ✓

$\Rightarrow \phi(n)$ ✓

Knowing $\phi(n)$ → find p, q

$(p-1)(q-1) = \phi(n)$

$$(p-1)\left(\frac{n}{p}-1\right) = \phi(n)$$

$$\Rightarrow \quad p^2 - \cdots p + \cdots = 0$$

$\Rightarrow$ Solve the quadratic to get p

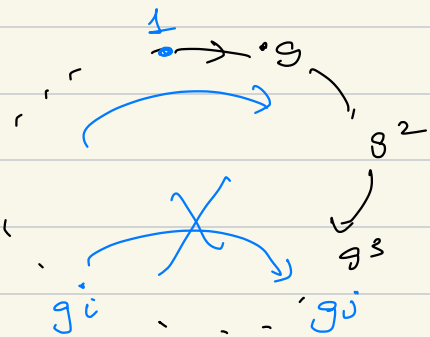
$$q = n/p$$

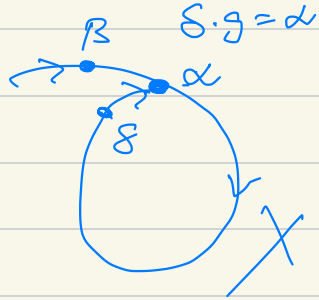
$G =$

$$\{g, g^2, g^3, \ldots g^{n-1}, 1\}$$

$g^n = 1$

$|G| = n$





Cyclic Groups

On certain Groups:

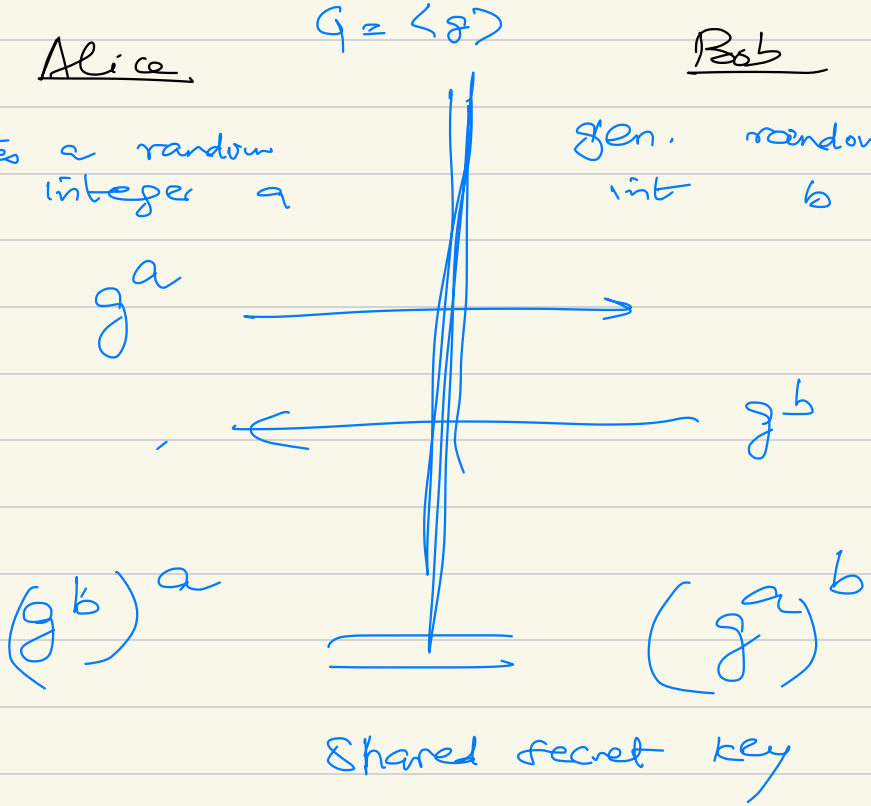
given (g, g^i) → computing i is hard

Discrete Log Problem (DLP)

Key Exchange

$G = \langle g \rangle$

$= \{g, g^2, \cdots g^{\cancel{?}}\}$



Diffie-Hellman Key Exchange

Public key Encryptions – RSA, El-Gamal, ...

Basic of Security

RSA → the RSA problem → depends on Factoring ($n = p \cdot q$)

El-Gamal → DLP

1997: Peter Shor : SIAM Jour. of Computing

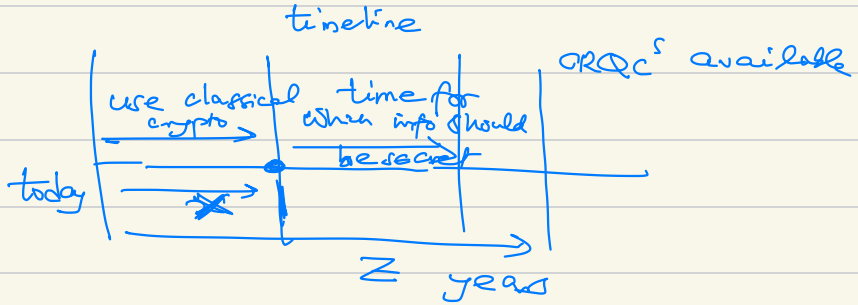
- Polynomial time algorithm for prime factorization and discrete log on a quantum computer

Today we have experimental/ very small sized quantum computer available (few qbits to few hundred qbits)

CRQC - Cryptographically Relevant Quantum Computer

- few million qbits, error tolerant

Mosca (2018)



if $(x+y) > z \Rightarrow$ Problem

Standards body NIST: PQC

Post-Quantum Crypto

Secure even in the presence of quantum comp.

Shor's algorithm:

finding period of a function

Periodic function:

$$f(x+T) = f(x) \quad \forall x$$

Shor: given $f(x) \rightarrow$ finds T

poly time algorithm

Factorization:

Goal: factor N.

(where $N = P \cdot Q$)

$$f(x) = a^x \bmod N \qquad \text{(randomly chosen } a\text{)}$$

$$\Rightarrow a^x \bmod N = a^{x+T} \bmod N$$

$$\Rightarrow a^T \bmod N = 1$$

assume T is even $= 2r$ (if not, then repeat with different 'a')

$$\Rightarrow a^{2r} \bmod N = 1$$

$$\Rightarrow (a^{2r} - 1) \bmod N = 0$$

$$\Rightarrow (a^r - 1)(a^r + 1) \equiv 0 \bmod N$$

$\gcd(a^r \pm 1, N) \longrightarrow$ factor of N.

$\Rightarrow$ Factorization of N is known

Period finding

- require a large no of $|0\rangle$ bits
- Quantum fourier Transform

Currently, we have factored $21 = 3 \times 7$

35

A few days ago (last week)

RSA-260 was solved

260 decimal digits

$\simeq$ 800 binary digits

Discrete Log Problem:

$G = \langle g \rangle$

Goal: given $(g, \underbrace{g^s}_{=y})$ find s

Define: $f(x_1, x_2) = g^{x_1} \cdot y^{x_2}$

Find period of this function $= (t_1, t_2)$

$$f(x_1+t_1, x_2+t_2) = f(x_1, x_2)$$

$\Rightarrow$

$$g^{x_1+t_1} \cdot y^{x_2+t_2} = g^{x_1} \cdot y^{x_2}$$

$\Rightarrow$

$$g^{t_1} \cdot y^{t_2} = 1$$

$\Rightarrow$

$$g^{t_1} \cdot g^{s \cdot t_2} = 1$$

$\Rightarrow$

$$g^{t_1 + s \cdot t_2} = 1$$

$\Rightarrow$

$$t_1 + s \cdot t_2 = 0 \quad \text{mod size of the group}$$

$\Rightarrow$

$$s = t_2^{-1} \cdot (-t_1).$$

NIST – Competition to design PQC schemes

for Enc.:

↳ Something similar to key exchange

↳ then use AES

for Signatures.

↳ ...

2024 NIST Standardized some schemes.

Security of these schemes depends on some other hard problem.

What happens to Symmetric Crypto when using Quantum Computers?

$AES(k, m) = c$

Attacker gets (m, c)

Goal: find k

Search over a database

$S = \{x_1, x_2, \ldots x_n\}$

Input: x

tell if $x \in S$

Unstructured database

Cost of search = $O(n)$

Lov Grover

On Quantum computer

search = $O(\sqrt{n})$

AES-128: key size = 128 bits

brute force = 2^{128}

Grover search = 2^{64}

AE-256:

Grover search $\geq 2^{128}$

13th Sep 2026

Yesterday:

(i) Classical crypto algs for IND-CCA security relying RSA problem (factoring related to DLP)

(ii) Quantum algorithms (Shor, Grover)

(iii) Mosca's inequality — Harvest Now, decrypt Later

NIST started PQC competition in ~ 2016
→ Standardized some schemes in 2024
→ Some are still under the process of standardization

NIST - RSA, El-Gamal, DHKE
— deprecated from 2030
— Disallowed from 2035

Hard Problems which Quantum Computers can't solve?

Languages: $L \subseteq \{0,1\}^*$

P: $L \in P$ for which there exists a decider in time polynomial in the length of the input

if $x \in L \Rightarrow D(x) = 1$

if $x \notin L \Rightarrow D(x) = 0$

NP: $x \in NP$

if $\exists\ w$ s.t. $|w| = \text{poly}(x)$

there exists a verifier V polytime algo

$$V(x, w) = 1$$

Useful for Crypto

Karp Reduction: $L_1 \leq_p L_2$

if there is a poly time algorithm f

s.t. $x \in L_1 \iff f(x) \in L_2$

eg: (RSA) $\leq_p$ (FACTOR)

BPP: Bounded Error Probabilistic Poly Time

$L \in BPP$ if

$x \in L \Rightarrow Pr[Algo(x) = 1] \geq 2/3$

$x \notin L \Rightarrow Pr[Algo(x) = 1] \leq 1/3$

$2/3$, $1/3$ are arbitrary $\quad 1/2 + \epsilon, \quad 1/2 - \epsilon$

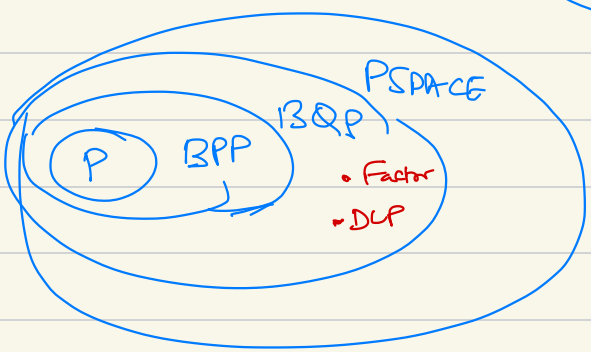
Succ = 0.7 → 70%

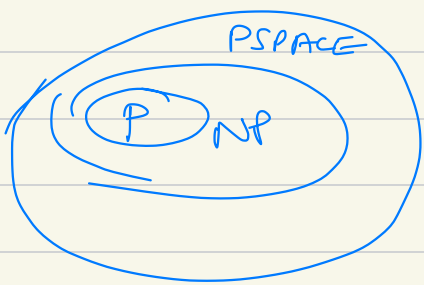
Three times success $= 1 - (0.3)^3 = 99.97\%$

BQP: Quantum version of BPP

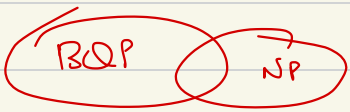
Both FACTOR & DLP lie in BQP

Ideally, we want problems in NP − BQP
(or NP \ BQP)





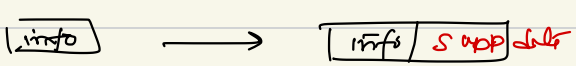
We don't know if BQP ⊆ NP or not

e.g. 

this COULD BE true

Various classes of problems of interest

① Oldest: Coding theory

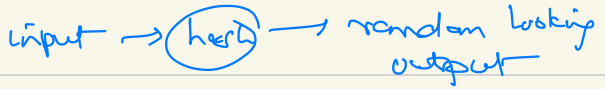


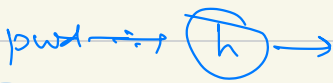
Sending only 1 bit → Repitition Code
1 1 1

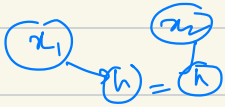
Receiver gets 1 0 1

McEliece Cryptosystem

(ii) Hash functions :







SHA-256 / 512

(iii) Lattice based – Lattice is an algebraic structure

Using certain hard problems on Lattices

(iv) multivariate equations

(v) isogeny mapping from $E_1 \not\to E_2$

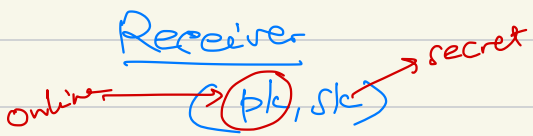
SIKE → about to be standardized

– Broken by 2 EU PhD students

– People are now working on such designs

(vi) some various types. . . –

~~Key-Exchange~~ Key-Encapsulation Mechanism (KEM)



Where C = ciphertext, K = secret key

Key Encapsulation Mechanism (KEM)

Key Decapsulation " (DEM)

use K to communicate

Security Goal: IND-CCA

C should be satisfying IND-CCA level of security

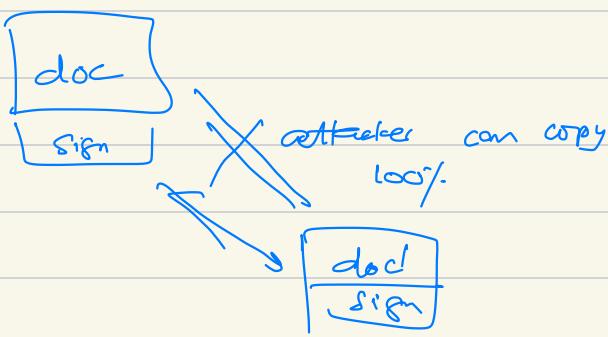
Signature:

Paper sign:



Assumption — sign. can't be faked or reproduced

Digital world



Signature should depend on

(i) the signer

& (ii) the document

How do people verify? — anyone should be able to run

Signing

(secret key, document)

↓

Sign

Verification

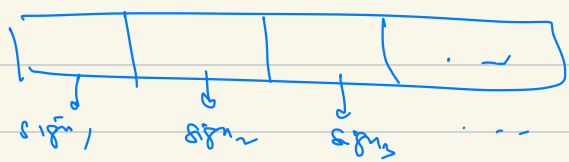
(pk, document, sign)

↓

true / false

RSA-2048 at its fastest implementation is about 100 times slower than AES

document:



In reality (for efficiency)

hash(document) = digest

Sign(digest)

You can't find 2 docs

